

LLM & RAG Evaluation — Master Guide

Beginner → Advanced | Knowledge + Interview Ready | Simple Examples

1. WHY EVALUATE LLMs?

LLMs are probabilistic. They can sound confident, fluent and professional while being wrong. Evaluation gives us a systematic way to measure quality instead of relying on “it looks good”.

Think of an LLM like a student:

Model = student; prompt = exam instructions; evaluation dataset = question paper; reference answer = answer key; metrics = marks; human evaluation = teacher review.

We evaluate: correctness, relevance, faithfulness, instruction following, safety, consistency, latency, cost and regressions.

Example: A medical RAG system answers “take the medicine every 2 hours”, but its source says every 6 hours. The answer sounds good but fails factual/grounded evaluation.

2. WHAT EXACTLY ARE WE EVALUATING?

Do not treat an LLM application as one black box.

Normal LLM:

Input → Prompt → LLM → Answer

Evaluate: correctness, relevance, instruction following, format, safety, latency and cost.

RAG:

Question → Retriever → Context → LLM → Answer

Evaluate separately:

1. Retrieval: did we find useful evidence?
2. Context: was the retrieved evidence relevant?
3. Generation: did the model use evidence correctly?
4. Final answer: did it answer the question?

Example: If the correct policy chunk was never retrieved, changing the generation prompt may not solve the real problem.

3. LLM EVALUATION

LLM evaluation measures the quality of model/application outputs.

Important dimensions:

- correctness
- relevance
- completeness
- instruction following
- factuality

- safety
- structured-output validity
- consistency

Example: Prompt says “return only one label”. Model returns “positive, because the review is good”. Classification may be correct, but instruction following failed.

For objective tasks use reference answers when possible. For open-ended tasks use rubrics, LLM judges and/or humans.

4. HALLUCINATION EVALUATION

Hallucination = unsupported, fabricated or incorrect information.

Example:

Context: “The document was written by Ritu.”

Answer: “The document was written by Rahul.” → hallucination.

For RAG, a practical approach is claim checking:

Answer → split into claims → find evidence → supported/unsupported → aggregate score.

Example:

Context: “Company founded in 2010.”

Answer: “Founded in 2010 and has 50 offices.”

First claim is supported; the second is unsupported if no evidence exists.

Important distinction: hallucination/factuality asks whether claims are true/supported; faithfulness specifically asks whether the answer is supported by the supplied context/evidence.

5. RAG EVALUATION

RAG evaluation checks retrieval AND generation.

Question → Retrieval → Context → Generation → Answer.

Three core questions:

1. Did we retrieve the right evidence?
2. Is the answer supported by that evidence?
3. Does the answer actually answer the user?

Example: For “What is the refund period?”, retrieving the refund policy is good retrieval. Answering “30 days” from that policy is good grounded generation. Retrieving company history instead is a retrieval failure.

6. RETRIEVAL METRICS

Precision@K = relevant retrieved / K.

Question: “How many of my top K results are relevant?”

Recall@K = relevant retrieved / total relevant items.

Question: “How many of all relevant items did I find?”

Hit Rate@K = 1 if at least one relevant result appears in top K, otherwise 0.

MRR = Mean Reciprocal Rank. It rewards the first relevant result being high:
rank 1 $\rightarrow$ 1.0; rank 2 $\rightarrow$ 0.5; rank 4 $\rightarrow$ 0.25.

nDCG = ranking metric for graded relevance. It rewards highly relevant documents appearing earlier.

Easy memory:

Precision = don't return too much junk.

Recall = don't miss useful information.

Hit Rate = did we find at least one useful thing?

MRR = how early did we find the first useful thing?

nDCG = how good is the whole ranking when relevance has levels?

7. RETRIEVAL DRY RUN

Question: "What is the refund period?"

Ground-truth relevant chunks = A and C.

Top 4 retrieved = B(irrelevant), A(relevant), D(irrelevant), C(relevant).

Precision@4 = $2/4 = 0.50$.

Recall@4 = $2/2 = 1.00$.

Hit@4 = 1.

MRR = $1/2 = 0.50$ because the first relevant result is rank 2.

Interpretation: we found all relevant chunks (excellent recall), but the ranking contains noise and the first useful result was not rank 1.

8. GENERATION METRICS

Generation evaluation checks the final answer.

Common dimensions: correctness, relevance, faithfulness, completeness, fluency and instruction following.

Traditional overlap metrics include BLEU and ROUGE. They compare generated text with a reference, but wording can differ while meaning remains correct.

Example:

Reference: "Paris is the capital of France."

Answer: "The capital city of France is Paris."

Meaning is correct even though wording differs.

Therefore modern evaluation often combines references, semantic/rubric evaluation, LLM judges and human evaluation.

9. FAITHFULNESS

Faithfulness asks: "Is the answer supported by the supplied context?"

Context: "Python was created by Guido van Rossum."

Answer: "Python was created by Guido van Rossum." $\rightarrow$ faithful.

If answer adds unsupported claims, faithfulness decreases.

Faithfulness is especially important for RAG because the purpose of retrieval is to ground generation in evidence.

Interview distinction:

Faithfulness $\neq$ general truth.

It specifically checks support from the evidence/context being evaluated.

10. ANSWER RELEVANCE

Answer relevance asks: "Does the answer address the user's question?"

Question: "What is the refund period?"

Answer: "The company was founded in 2015 and has 20 offices."

Even if true, it is irrelevant.

Memory:

Faithfulness = Is the answer supported?

Answer relevance = Does it answer the user?

11. CONTEXT RELEVANCE

Context relevance asks: "Are the retrieved contexts useful for this question?"

Question: "What is the refund period?"

Retrieved: refund policy, company history, office locations.

Refund policy = relevant.

History/locations = irrelevant.

This is mainly a retrieval-side quality concept.

Memory:

Context relevance $\rightarrow$ useful evidence?

Faithfulness $\rightarrow$ answer supported?

Answer relevance $\rightarrow$ question answered?

12. THE THREE CORE RAG QUESTIONS

Think of RAG quality as a triangle:

Retrieval $\rightarrow$ Did we get the right context?

Faithfulness $\rightarrow$ Is the answer supported by that context?

Answer relevance $\rightarrow$ Does the answer answer the question?

A system can fail at any one:

- right question + wrong retrieval
- right context + hallucinated answer
- supported answer + answers the wrong question

13. RAGAS

RAGAS is a framework for evaluating RAG pipelines with metrics covering aspects such as faithfulness, answer relevance and retrieval/context quality, depending on the configured metrics/version.

Conceptual flow:

Question + Context + Answer (+ Reference when available) → RAGAS metrics → report.

Example:

Question: "What is the refund period?"

Context: refund-policy chunks.

Answer: "30 days."

Reference: "Refunds can be requested within 30 days."

RAGAS helps evaluate separate parts of RAG rather than only saying "the final answer looks good".

Interview point: RAGAS is not a magical ground-truth oracle. Scores depend on the dataset, metric definition, evaluator model, prompts and implementation/version. Use it with task-specific tests and human evaluation where appropriate.

14. LLM-AS-A-JUDGE

LLM-as-a-Judge means using an LLM to evaluate another model's output with a rubric.

Example rubric:

Correctness 0–5

Relevance 0–5

Completeness 0–5

Advantages: scalable, fast, useful for open-ended responses.

Risks: judge bias, verbosity/style bias, position bias, model-family bias, poor calibration and judge hallucination.

Pointwise judging = score one answer.

Pairwise judging = compare Answer A vs Answer B.

Best practice: clear rubric, structured judge output, representative examples, and calibration against human labels. High-stakes systems should not rely only on an LLM judge.

15. EVALUATION DATASETS

An evaluation dataset is a collection of test cases used to measure quality.

For RAG it can contain:

question, reference answer, relevant document/chunk IDs, retrieved contexts, generated answer, human label and metadata.

Include:

- easy + normal cases
- difficult/ambiguous cases
- multi-hop cases

- out-of-scope questions
- adversarial/prompt-injection cases
- negative cases
- long-context cases

Golden dataset = carefully reviewed trusted set.

Regression dataset = previous failures that must continue to pass.

Avoid train/evaluation leakage: if you tune your system directly on test examples, the score becomes misleading.

16. HUMAN EVALUATION

Humans evaluate outputs using a rubric.

Example 1–5 helpfulness rubric:

5 = correct, direct, complete, no unsupported claims.

3 = mostly correct but missing an important point.

1 = incorrect or fails to answer.

Good practice:

- clear rubric
- calibration examples
- representative sample
- multiple annotators for difficult/high-stakes tasks
- measure agreement
- blind model identity where practical
- adjudicate disagreements

Human evaluation is expensive, so it is often used to validate automated evaluation rather than inspect every response.

17. OFFLINE VS ONLINE EVALUATION

Offline evaluation = fixed dataset before deployment or during development.

Flow:

Version → evaluation dataset → metrics → compare → deploy.

Useful for prompt changes, model comparison, retriever tuning and regression testing.

Online evaluation = real production behavior.

Examples:

user feedback, sampled human review, production quality metrics, latency, tool failures, escalation rate, online judge.

Easy memory:

Offline = controlled laboratory test.

Online = real-world monitoring.

Use both.

18. CONTINUOUS EVALUATION LOOP

Build → offline evaluation → improve → deploy → online monitoring → collect failures → add failures to regression dataset → evaluate again.

A mature LLM system turns production failures into future tests.

This is one of the strongest interview points because evaluation is not a one-time activity.

19. COMPLETE RAG EVALUATION EXAMPLE

Medical RAG question: “What are the symptoms of condition X?”

1. Retrieval: Did we retrieve the correct medical chunks? Use Recall@K, Precision@K, Hit Rate, MRR/nDCG.
2. Context: Are those chunks relevant?
3. Generation: Is the answer supported? Use faithfulness.
4. User alignment: Does it answer the question? Use answer relevance.
5. Overall correctness: compare with trusted reference/human rubric where available.
6. Safety: check unsupported medical claims and escalation requirements.
7. Production: monitor latency, cost, failures and user feedback.

Therefore “RAG accuracy” is not one single metric.

20. HOW TO DEBUG A BAD RAG SYSTEM

Case A: Recall@5 is low → retrieval misses evidence.

Try better embeddings, hybrid retrieval, query rewriting, chunking, reranking, metadata filters or larger K.

Case B: retrieval is good but faithfulness is low → generation is not staying grounded.

Investigate context formatting, grounding instructions, answerability checks, model choice and output constraints.

Case C: faithfulness high but answer relevance low → answer is supported but doesn't address intent.

Improve query understanding, prompt and output criteria.

Case D: offline metrics high but users complain → dataset may not represent production.

Collect real failures, segment traffic and add cases to evaluation.

21. MOST IMPORTANT INTERVIEW DEBUGGING QUESTION

Question: “My RAG answer is wrong. What do you do?”

Strong answer:

“I first separate retrieval from generation. I inspect retrieved chunks and calculate retrieval metrics such as Recall@K, Precision@K, Hit Rate, MRR or nDCG. If the correct evidence was not retrieved, I improve retrieval. If the correct evidence was retrieved but the answer is unsupported or incorrect, I investigate faithfulness, answer relevance and generation. I avoid changing the prompt blindly before locating the failing component.”

22. METRIC COMPARISON

Precision@K → How much of top-K is relevant?

Recall@K → How much relevant information was found?

Hit Rate@K → Did top-K contain at least one relevant result?

MRR → How early is the first relevant result?

nDCG → How good is the ranked list with graded relevance?

Context relevance → Is retrieved context useful?

Faithfulness → Is answer supported by context?

Answer relevance → Does answer address the question?

RAGAS → Framework for RAG evaluation.

LLM-as-Judge → LLM scores output using a rubric.

Human evaluation → People judge quality.

Offline → fixed controlled testing.

Online → production monitoring.

23. TOP INTERVIEW QUESTIONS

Q1 Why evaluate LLMs? Because outputs are probabilistic and can be fluent but wrong, irrelevant, unsafe or inconsistent.

Q2 What do you evaluate? Correctness, relevance, instruction following, safety, format, latency/cost; for RAG also retrieval, grounding and final answer quality.

Q3 What is hallucination? Unsupported/fabricated/incorrect generated information.

Q4 What is RAG evaluation? Evaluation of both retrieval and generation.

Q5 Precision vs Recall? Precision measures how much retrieved content is relevant; recall measures how much of the relevant content was retrieved.

Q6 Hit Rate? Whether at least one relevant item is in top K.

Q7 MRR? Mean Reciprocal Rank; rewards the first relevant result being near the top.

Q8 nDCG? Ranking metric supporting graded relevance and position.

Q9 Faithfulness? Whether answer is supported by provided evidence.

Q10 Faithfulness vs answer relevance? Supported vs actually answers the question.

Q11 Context relevance? Whether retrieved context is relevant/useful.

Q12 RAGAS? A RAG evaluation framework with configurable metrics for aspects such as faithfulness, relevance and retrieval/context quality.

Q13 LLM-as-a-Judge? An LLM evaluates outputs using a rubric.

Q14 Why not only use an LLM judge? Bias/calibration limitations; validate against humans.

Q15 Golden dataset? Trusted, reviewed evaluation set.

Q16 Offline vs online? Fixed controlled test set vs real production behavior.

Q17 Why human evaluation? Some nuanced or high-stakes quality dimensions are hard to automate.

Q18 How debug wrong RAG? First determine retrieval failure vs generation/grounding failure.

24. ADVANCED INTERVIEW QUESTIONS

Q1: Recall@10 high but answers wrong. Why?

Retrieval may contain evidence but ranking/context may be noisy, the generator may ignore evidence, or documents may conflict. Inspect context relevance and faithfulness.

Q2: Precision high but recall low?

Retrieved results are mostly relevant, but useful evidence is being missed. Improve recall with retrieval/query/chunking methods.

Q3: Why not maximize recall blindly?

Too much irrelevant context increases noise, cost, latency and can hurt generation.

Q4: Why can retrieval improve while final answers worsen?

Retrieval metrics are component-level; more retrieved content can add noise. Optimize end-to-end quality too.

Q5: Why can offline evaluation mislead?

Dataset may be too small/easy, stale, leaked, or different from real traffic.

Q6: How build a good dataset?

Use representative intents, edge/adversarial/out-of-scope cases, trusted references/relevance labels, versioning and regression failures.

Q7: How evaluate hallucination without reference answers?

Use context-grounded claim checking, human review or a validated judge/rubric depending on the task.

Q8: What is distribution shift?

Production query distribution differs from development/evaluation data.

Q9: Why add production failures to the dataset?

They become regression tests and reduce repeated failures.

25. PRACTICAL CHECKLIST

Before deployment:

- ☐ Define "good".
- ☐ Separate retrieval and generation.
- ☐ Build representative evaluation data.
- ☐ Add edge/adversarial/out-of-scope cases.
- ☐ Define retrieval metrics.
- ☐ Define grounding/relevance metrics.
- ☐ Add task-specific correctness checks.
- ☐ Validate LLM judge against humans.
- ☐ Run offline regression tests.
- ☐ Set thresholds.
- ☐ Compare versions.
- ☐ Monitor latency/cost.

- Monitor production quality.
- Collect user feedback.
- Turn failures into regression tests.

26. FINAL MASTER MENTAL MODEL

WHY? → LLMs can sound right while being wrong.

WHAT? → correctness, relevance, safety, instruction following, structure, latency, cost.

RAG:

Question → Retrieval → Context → Generation → Answer.

Evaluate:

Retrieval → Precision/Recall/Hit Rate/MRR/nDCG

Context → Context Relevance

Grounding → Faithfulness

User alignment → Answer Relevance

Overall → task-specific correctness/reference/human rubric

Scalable judge → LLM-as-a-Judge

Framework → RAGAS

Human quality → Human Evaluation

Development → Offline Evaluation

Production → Online Evaluation

Golden interview principle:

DO NOT immediately change the LLM or prompt.

First identify which component failed:

retrieval failure → fix retrieval;

grounding failure → fix generation;

relevance failure → fix intent/output;

offline-good/online-bad → improve dataset and monitoring.